



OPTIGA™ TPM SLB9672

A future-proof new generation TPM

Infineon Technologies
September 2023



Infineon's award-winning TPM technology



Several awards testify to the innovative strengths and advanced cryptographic capabilities of

Our OPTIGA™ TPM SLB 9672/9673 solutions

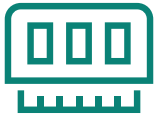
“Embedded Award 2023” from Embedded World
First place in the “Safety&Security” category

“Best in Show” award from Embedded Computing Design
Top spot in the “Security” category

Product of the Year” award from ELEKTRONIK
First prize in the “Software” category



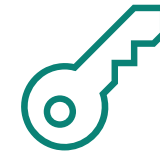
Why security is essential



Security is a fundamental need of society with increasing importance



The connected world is further driving the demand for security



We believe in hardware-based security as the essential trust anchor

TPM as Root of Trust

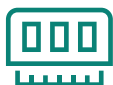


Discrete TPM, key Root-of-Trust for multiple applications

Key targets of discrete TPM

PC & laptops

- Professional PCs
- Industrial PCs



Servers

Servers



IoT networking

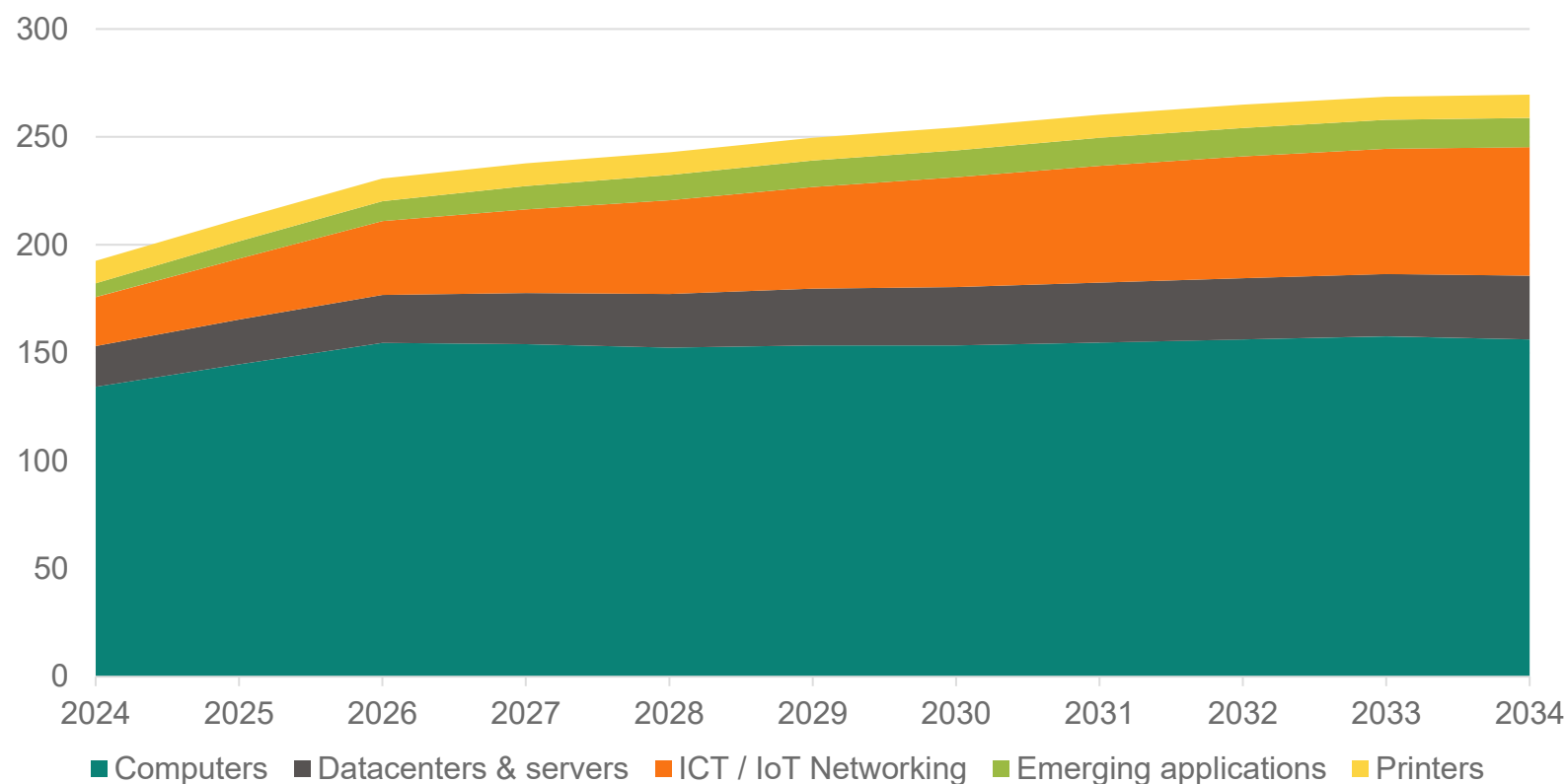
- Network Interface Cards
- Networking equipment
- Printers



Forecasted markets for discrete TPM

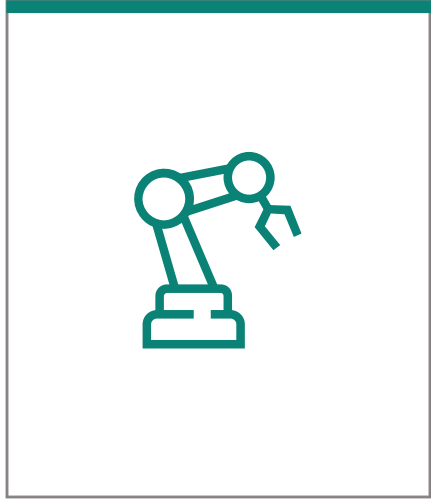
A stable base market and significant growth in other segments

Market size (in M pcs)

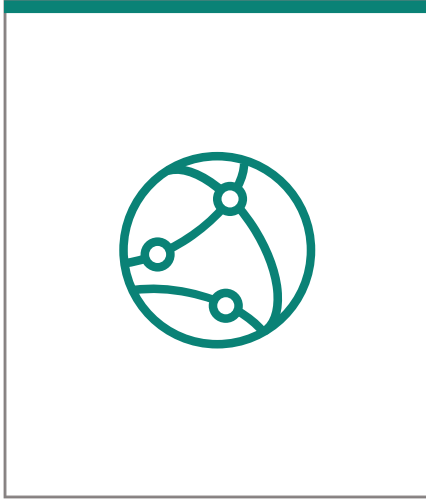


What a TPM does

Smart factory



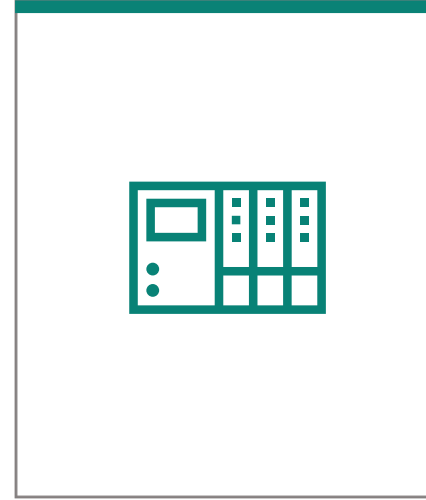
IoT networking



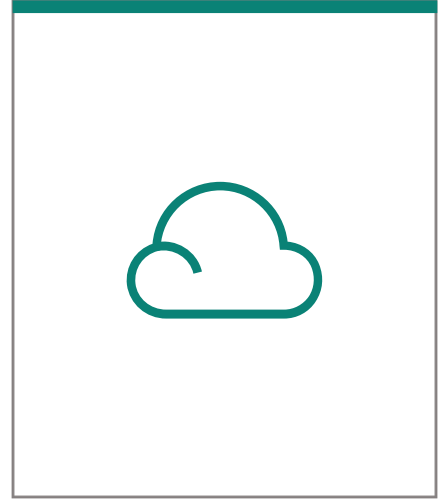
PC & laptops



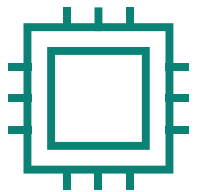
Servers



Cloud



- Offers a standardized solution
- Allows trust and secured communications
- Allows protection of exchanged valuable data
- Supports the latest security requirements
- Is updatable



Future challenges for TPM

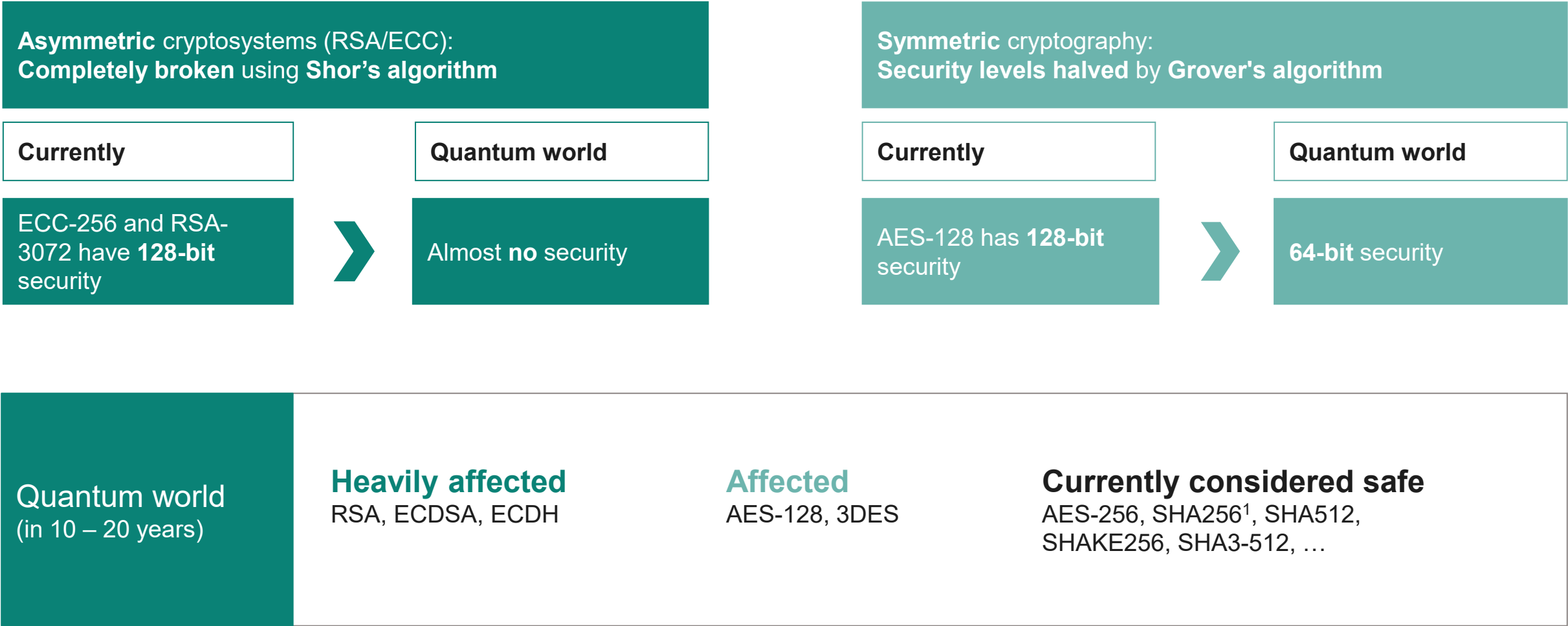


The threat of quantum computers to cryptography

Within the next 10 to 20 years, quantum computer attacks on today's cryptography are expected to become reality.



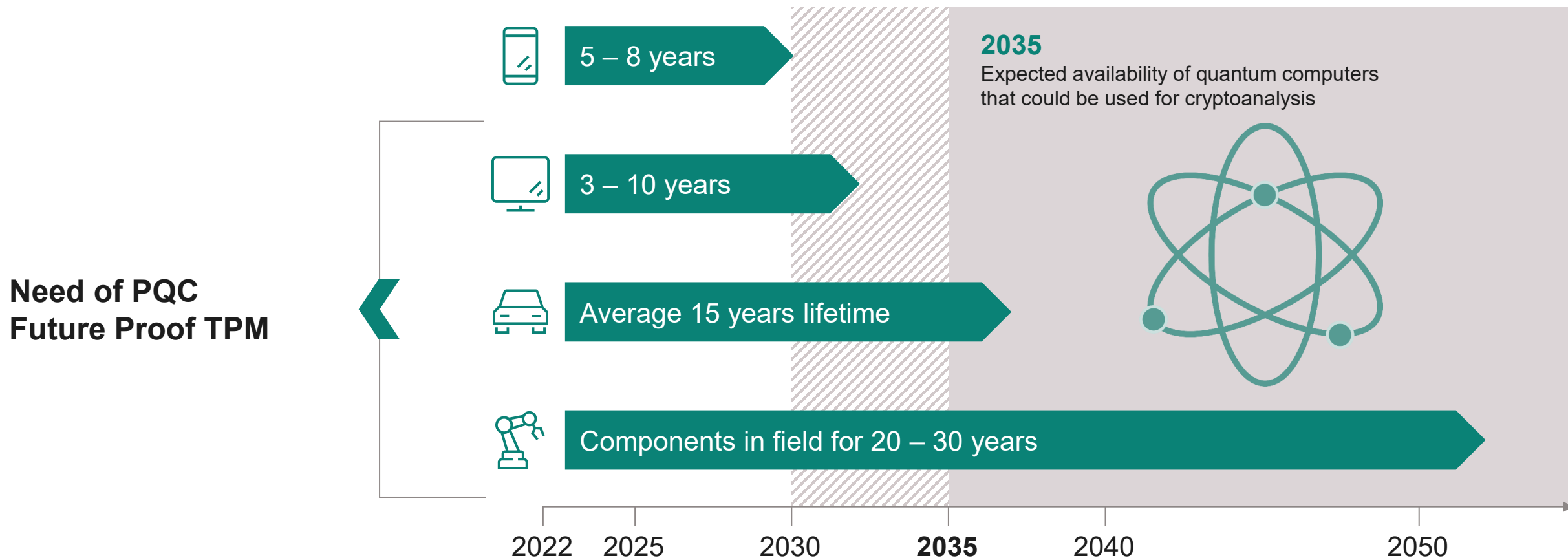
Quantum computers, a threat to currently known security algorithms



¹ Preimage resistance

Considered timeline

Devices with over 10 years lifecycle must be prepared for the quantum computing age



The security of TPM applications can only be as high as the one of the firmware update mechanism



In the past

Embedded device

Firmware update mechanism

128-bit classical security

Embedded application
128-bit classical security



Today

Embedded device

Firmware update mechanism

128-bit PQC security

Embedded application
128-bit (or more) classical security



Use HBS standards available today



In the near future

Embedded device

Firmware update mechanism

128-bit PQC security

Embedded application
128-bit PQC security



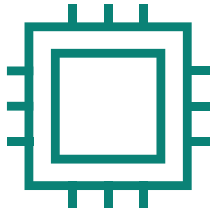
Upgrade to future
PQC standards

OPTIGA™ TPM

SLB 9672



Infinion has already taken the first steps into the world of quantum computing



OPTIGA™ TPM SLB 9672

The first TPM on the market with a **PQC-protected** firmware update mechanism



The key benefits with Infineon's newest TPM family member



Future-proof

- PQC-protected firmware update mechanism
- Extended memory
- Stronger cryptographic algorithms



Robust security

- Improved computational performances
- Resiliency features
- Fully compliant with the TCG requirements and certified accordingly



Easy integration

- Standardized Root of Trust
- Tools to support design activities
- Supports the latest version of Windows and Linux

OPTIGA™ TPM SLB 9672, a future-proof TPM

Previous generation TPM

Firmware update



ECDSA



TCG certified Version 2
As per Revision 1.38

Improvements

OPTIGA™ TPM SLB 9672

Firmware update



ECDSA



XMSS



New stronger
crypto algorithms

Resiliency
features

Quantum resistant

RSA 3k & 4k
SHA-384, ECC 384

To avoid any risk
of FW corruption

TCG certified Version 2
As per Revision **1.59**

One device – Two solutions

Firmware version



Standard Edition (FW 15.2x)

Optimized for PCs, servers

Enhanced Security for IoT networking (FW 16.1x)

Functionalities & applications



Primary choice for MSFT Windows environment/ecosystem
and connected devices
with a “PC platform” architecture.

- Two product variants:
- Standard temperature range -20° C to +85° C
- Extended temperature range -40° C to +85° C

Suitable for connected devices supporting enhanced security features

- Chip Unique ID readout
- AES encryption and decryption
- Disabling EK key deletion

Two product variants:

- Extended temperature range for -40° C to +85° C
- Extended temperature range for -40° C to +105° C

The benefits of a hardware-based security



Why hardware-based security?



No security

Open for all to see



Software security only



Hardware security

Reading

Software code easily readable by attackers

Hardware chip protects itself against code reading

Copying

Software code easily copied and shared by attackers

Security hardware must be reverse engineered and re-manufactured

Analyzing

Software code easily analyzed and understood using standard tools

Hardware protection for data processing, transport storage

Root of Trust

Consequently, not so strong “Root of Trust” anchor for the system

Strong “Root of Trust” anchor for the system, providing detection, recoverability, secured updates

Relying on Infineon's hardware-based security protects secret keys against software vulnerabilities in OS and Apps

Why software security is often not enough?



Secret keys kept in the shared memory



Secret keys securely kept in the OPTIGA™ TPM

Security adds value by protecting your business, enabling growth and saving costs

Protecting

- Trust and reputation
- IP and process know-how
- Long-term revenue & profitability of investments



Enabling

- Growth
- New business models
- Security as a differentiation factor



Saving

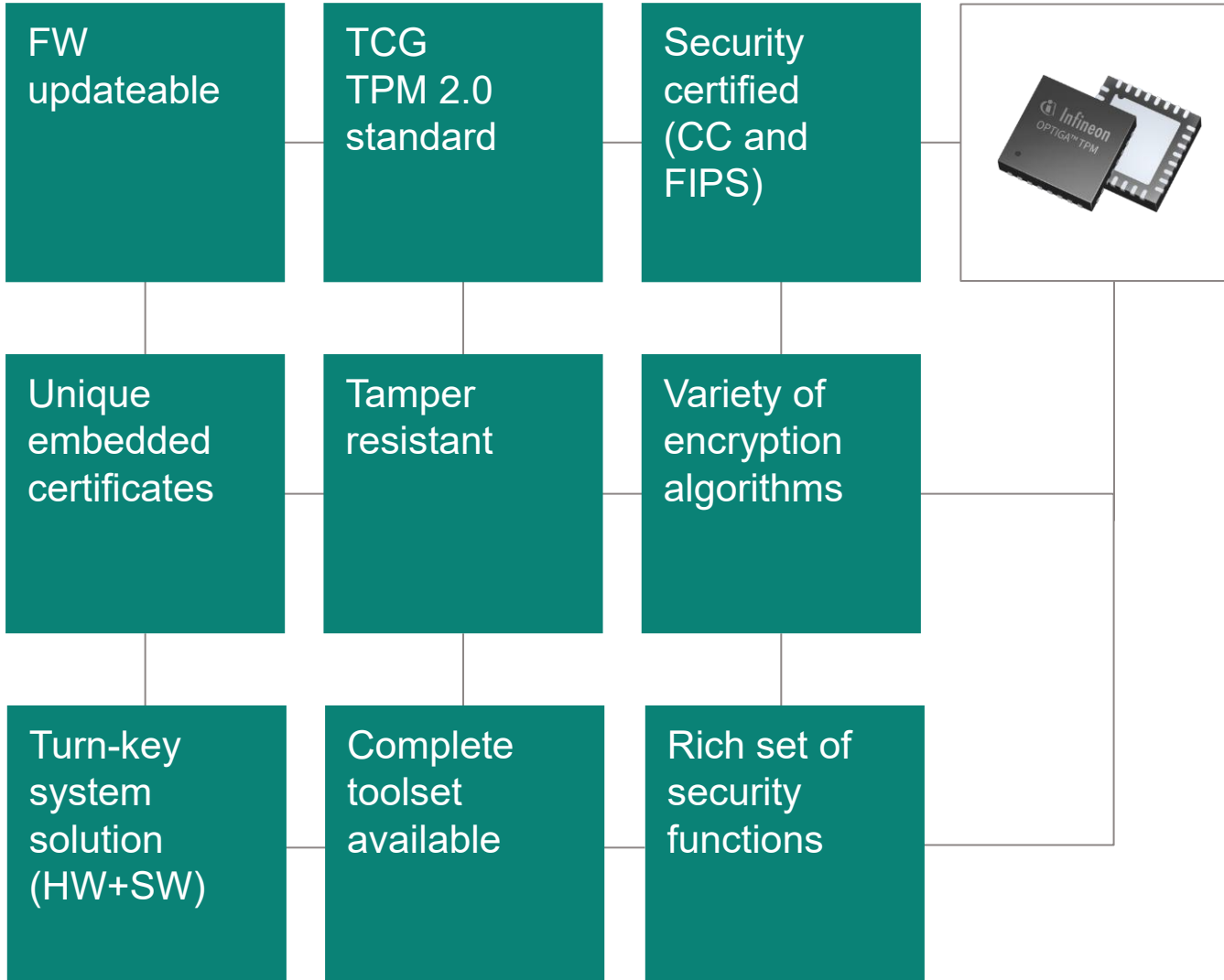
- Costs by preventing security-related system interruptions
- Cost based on new ways of solving a problem



Why the OPTIGA™ TPM family



Every second business laptop comes with an OPTIGA™ TPM



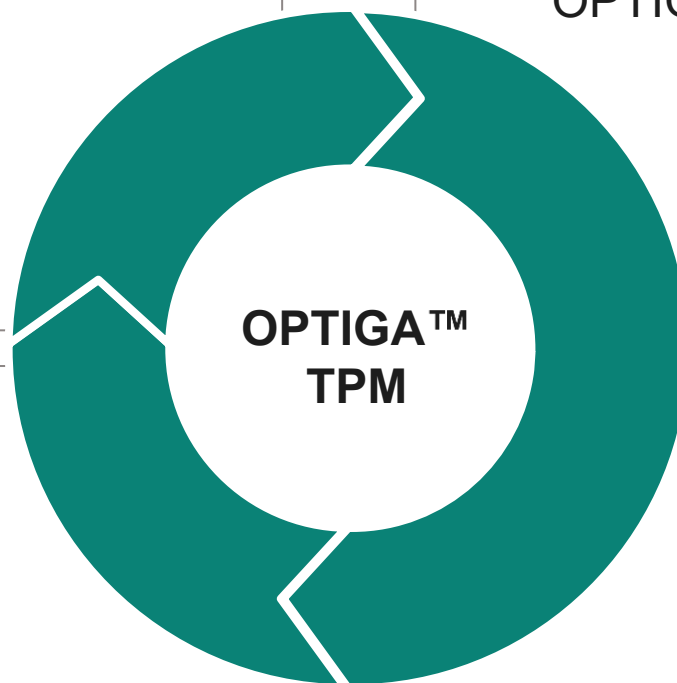
OPTIGA™ TPM family offers rich functionality and flexibility



OPTIGA™ TPM SLM 9670
Industrial



OPTIGA™ TPM
SLI 9670
inCar



OPTIGA™ TPM SLB 9670
OPTIGA™ TPM SLB 9672
OPTIGA™ TPM SLB 9673
Consumer/IoT



Our solution comes with service and support

We support you by ...



- Providing Design-In Application Notes for our Products
- Host side integration support
- Evaluation Kits



- Providing a secured Public Key Infrastructure
- Custom certificate loading in secured Production Environment



- Answering questions immediately
- Two Level Customer service



- Providing trainings for our security products
- Showing Demo Applications as a starting point for custom designs



Key take-aways

Security ...

... is essential and HW-based security provides benefits beyond strong security including time to market, logistics and scalability



New requirements ...

... coming in near the future because of quantum computers and the threat to existing cryptographic algorithms



OPTIGA™ TPM SLB 9672 ...

... is the right choice if you want to meet the challenges of today and tomorrow



Information and tools for OPTIGA™ TPM
are easily available on Infineon's website

www.infineon.com/tpm

and our Github repository

<https://github.com/Infineon/OPTIGA-TPM>



